# Assignment 4: Semantics I

Ling 201 – Fall 2026

*(100 points total)*

Please **do not use external sources such as the internet or AI to find the answers**. If you need an **extension**, please ask before the deadline.

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### Part 1. Truth Values *[difficulty level: ☆]*

Consider the following two worlds. Then answer the question below about these possible worlds.

| **World A** | | | **World B** | | |
|---|---|---|---|---|---|
| happy | → | {Quartz, Utku, Junwoo, Sukchan} | happy | → | {Junwoo, Tetsu, Chenli, Fahad} |
| tired | → | {Vincent, Chenli, Fahad} | tired | → | {Quartz, Merlin} |
| friendly | → | {Utku, Junwoo, Chenli, Beryl} | friendly | → | {Sukchan, Vincent, Beryl, Merlin} |
| brave | → | {Merlin, Sukchan} | brave | → | {Utku, Junwoo} |
| running | → | {Quartz, Utku, Chenli} | running | → | {Utku, Vincent, Beryl} |
| sleeping | → | {Vincent, Fahad} | sleeping | → | {Quartz, Junwoo, Sukchan} |
| reading | → | {Junwoo, Tetsu, Merlin} | reading | → | {Chenli, Fahad, Merlin} |
| singing | → | {Sukchan, Beryl, Chenli} | singing | → | {Utku, Tetsu, Beryl} |

For each statement below, **indicate** whether it would be true or false in World A and World B, as defined above. *(1 point/statement, 20 points total)*

| **Statement** | **World A** | **World B** |
|---|---|---|
| 1. Quartz is running. | | |
| 2. Vincent is not sleeping. | | |
| 3. Junwoo is a friendly reader. | | |
| 4. Fahad is a tired singer. | | |
| 5. Sukchan is brave and singing. | | |
| 6. Chenli is happy and running. | | |
| 7. Merlin is friendly but not tired. | | |
| 8. Beryl is singing and friendly. | | |
| 9. Utku is a tired reader. | | |
| 10. Tetsu is sleeping and happy. | | |

**Part 2. Entailment Relations I *[difficulty level: ☆]***

For each pair of sentences below, determine **whether or not sentence (a) entails sentence (b)**, and **explain briefly why** in your own words. *(5 points/item, 30 points total)*

1. (a) Mount Everest is taller than every other mountain.
   (b) Mount Everest is taller than Mount Kilimanjaro.

2. (a) Arthur went for a run at noon.
   (b) Arthur went for a run in the park at noon.

3. (a) Morgana said she studied for the test.
   (b) Morgana studied for the test.

4. (a) Lancelot ate nachos and enchiladas for dinner.
   (b) Lancelot ate enchiladas for dinner.

5. (a) Mordred played a song on his guitar.
   (b) Mordred played music.

6. (a) Guinevere read a book yesterday.
   (b) Guinevere read a novel yesterday.

**Part 3. Entailment Relations II *[difficulty level: ☆☆☆]***

For each pair of sentences below, identify the **entailment relations between two propositions** (*p* and *q*) and **justify your answer using semantic reasoning** (truth conditions, etc.).

**Possible entailment relations are:**

1) *p* entails *q*, but *q* does not entail *p*;

2) *q* entails *p* but *p* does not entail *q*;

3) *p* entails *q* and *q* entails *p* in the meantime (mutual entailment); and

4) *p* does not entail *q* and *q* does not entail *p*, either (no entailment relation between *p* and *q*).

**Example:**

*p*: I am very tired. *q*: I am tired.

This is an example of "*p* entails *q*, but *q* does not entail *p*".

i. 'I am very tired' also means 'I am tired'. Hence, if *p* is **TRUE**, *q* must also be **TRUE**.
ii. 'I am tired' does not mean 'I am very tired', since I might be 'slightly tired' only. Hence, if *q* is **TRUE**, *p* is not necessarily **TRUE**.

*(8 points/pair, 32 points in total)*

a) *p*: Uther solved all questions on the exam.
*q*: Uther solved some of the questions on the exam.

b) *p*: Merlin broke the window.
*q*: The window was broken by Merlin.

c) *p*: Gawain took the driving test.
*q*: Gawain failed the driving test.

d) *p*: Perceval is reading a novel
*q*: Perceval is writing an essay.

**Part 4. Identifying Entailment *[difficulty level: ☆]***

For each of the following sentences, **write down two** more sentences that are entailed by that sentence. *(3 points/item, 18 points total)*

1. Dungeons & Dragons, which was created in Wisconsin, is the most well-known tabletop role-playing game, and it has influenced video game genres and "geek" subculture since it came out in the 1970s.

2. Janine Teagues, a cheerful and optimistic second-grade teacher, helps her students learn spelling, reading, and other subjects every day at Abbott Elementary School in Philadelphia, despite limited resources and her boss Ava Coleman's erratic behavior.

3. *Epic: The Musical* is a nine-part series of small albums that recounts the Ancient Greek poem *The Odyssey*, which tells the story of Odysseus's long journey home to his wife and son after the Trojan War, with many exciting adventures along the way.